

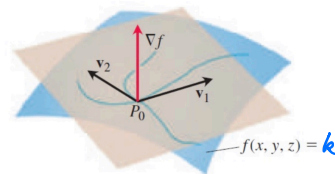
MAT1002 Lecture 16

Outline

- Tangent planes and linear approximation
- Optimization

Tangent Planes and Linear Approximation

- ▶ Let $\mathcal{S}$ be a surface in the domain of f described by $f(x, y, z) = k$, where f is differentiable.
- ▶ Let (x_0, y_0, z_0) be a point in $\mathcal{S}$.
- ▶ Take any curve $\mathcal{C}$ on $\mathcal{S}$ passing through (x_0, y_0, z_0) .
- ▶ Just like what happened to level curves, $\nabla f(x_0, y_0, z_0)$ is perpendicular to any tangent vector of $\mathcal{C}$ at (x_0, y_0, z_0) .



By considering different curves passing through the point, we have the picture above.

Def. Given any level surface $\mathcal{S}$ given by $f(x, y, z) = k$ of a differentiable function f , the **tangent plane** at a point $P_0 \in \mathcal{S}$ to $\mathcal{S}$ is the plane through P_0 with normal $\nabla f(P_0)$, and the **normal line** of the surface at P_0 is the line through P_0 in the direction of $\nabla f(P_0)$.

The **tangent plane** to a level surface of f at (x_0, y_0, z_0) is described by

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0.$$

In particular, for graph of $f(x, y)$ (surface $z = f(x, y)$), the tangent plane to it at $P_0 = (x_0, y_0, f(x_0, y_0))$ is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - f(x_0, y_0)) = 0$$

or equivalently,

$$Z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The function L whose graph is the tangent plane above, is given by

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The function L is called the **linearization** of f at (x_0, y_0) . The approximation

$f(x, y) \approx L(x, y)$,
 (standard) is referred to as the **linear approximation** or the **tangent plane approximation** of f at (x_0, y_0) .

Error of Approximation

Now fix a starting point (x_0, y_0) and consider the linear approximation. The **error** of the approximation is $E(x, y) := f(x, y) - L(x, y)$.

Suppose:

- $f_x, f_y, f_{xx}, f_{yy}, f_{xy}, f_{yx}$ are all continuous on an open region D .
- In some rectangle R centered at (x_0, y_0) , where $R \subseteq D$, $\exists M$ such that $|f_{xx}|, |f_{yy}|$ and $|f_{xy}| (= |f_{yx}|)$ are all bounded above by (the same) M .

Then

$$|E(x, y)| \leq \frac{1}{2}M(|x - x_0| + |y - y_0|)^2.$$

for all $(x, y) \in R$.

We will postpone the proof to a later section.

Differentials

Analogous to the one-variable case $y = f(x)$, we define differentials for a differentiable function $z = f(x, y)$ similarly. Given any point (x_0, y_0) in the domain of f :

- Introduce independent variables Δx and Δy .
- Also them as dx and dy ($dx = \Delta x$, $dy = \Delta y$).
- $\Delta z = \Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$. (Change in f)
- $dz = df = L(x_0 + dx, y_0 + dy) - L(x_0, y_0)$
 $= f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$. (Change in L)

This dz is called the total differential. Note that it is a function of x_0, y_0, dx and dy .

e.g. Consider a right circular cylinder with base radius r and height h .

$$\begin{aligned} \cdot V &= \pi r^2 h \\ \cdot dv &= \frac{\partial V}{\partial r} dr + \frac{\partial V}{\partial h} dh = 2\pi r h dr + \pi r^2 dh \end{aligned}$$

How much volume would change approximately if r and h are changed by "small" dr and dh , respectively.

Functions with Three Variables

For $w = f(x, y, z)$, differentiable, "base point" $P_0 = (x_0, y_0, z_0)$:

• Linearization of f at P_0 :

$$L(x, y, z) = f(P_0) + f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0)$$

• Error: If all second-order partials are cts on an open D and their absolute values are bounded by M on $\underline{R} \subseteq D$,
rectangular solid.
then

$$|E(x, y, z)| \leq \frac{1}{2} M (|x - x_0| + |y - y_0| + |z - z_0|)^2.$$

for all $(x, y, z) \in R$.

• Total differential: $dw = df = f_x(P_0)dx + f_y(P_0)dy + f_z(P_0)dz$.

e.g. For $f(x, y, z) = x^2 - xy + 3\sin z$, consider the standard linear approximation at $P_0 = (2, 1, 0)$ on

$$\begin{aligned} R &= [1.99, 2.01] \times [0.98, 1.02] \times [-0.01, 0.01] \\ &= \{(x, y, z) : x \in [1.99, 2.01], y \in [0.98, 1.02], z \in [-0.01, 0.01]\}. \end{aligned}$$

• $f(P_0) = 2$, $f_x(P_0) = 3$, $f_y(P_0) = -2$, $f_z(P_0) = 3$, so

$$L(x, y, z) = 2 + 3(x - 2) - 2(y - 1) + 3z = 3x - 2y + 3z - 2.$$

• Since $f_{xx} = 2$, $f_{yy} = 0$, $f_{zz} = -3\sin z$, $f_{xy} = -1$, $f_{xz} = 0$, $f_{yz} = 0$,

all their absolute values ≤ 2 .

Hence, for all (x, y, z) in R ,

$$\begin{aligned} |E(x, y, z)| &\leq \frac{1}{2} \cdot 2 (|x-2| + |y-1| + |z-0|)^2 \\ &\leq (0.01 + 0.02 + 0.01)^2 = 0.0016. \end{aligned}$$

Optimization

Local Extrema

Definition

Let $f : D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$, be a function, and let $(a_1, \dots, a_n)$ be a point in D .

- We say that f has a **local maximum** at $(a_1, \dots, a_n)$ if there exist an $r > 0$ such that

$$f(a_1, \dots, a_n) \geq f(x_1, \dots, x_n)$$

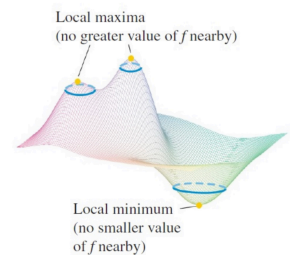
for every $(x_1, \dots, x_n) \in D \cap B_r(a_1, \dots, a_n)$.

open ball with radius r
(disk if $n=2$)

- We say that f has a **local minimum** at $(a_1, \dots, a_n)$ if there exist an $r > 0$ such that

$$f(a_1, \dots, a_n) \leq f(x_1, \dots, x_n)$$

for every $(x_1, \dots, x_n) \in D \cap B_r(a_1, \dots, a_n)$.



Caution: For $f(x, y) = x^2 + y^2$ defined on the closed disk

$$D = \{(x, y) : x^2 + y^2 \leq 1\},$$

f has local maxima at all (x, y) satisfying $x^2 + y^2 = 1$, the boundary of D .

Local extrema are also called **relative extrema**.

Although many theorems for **extrema** (i.e., maxima or minima) hold for n -variable functions, we will focus on these theorems for functions with 2 or 3 variables.

Theorem (First Derivative Test)

For a function $f : D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^2$, suppose that:

- ▶ f has a local maximum or a local minimum at an **interior point** (a, b) of D , and;
- ▶ both $f_x(a, b)$ and $f_y(a, b)$ exist

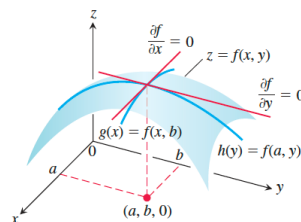
Then $f_x(a, b) = 0$ and $f_y(a, b) = 0$. That is, $\nabla f(a, b) = \vec{0}$.

Proof: • Define $g(x) := f(x, b)$. Then $g(a)$ is a local extremum, where a is an interior point of the domain of g . By one-variable theory, $g'(a)$ D.N.E or $g'(a) = 0$.

• $g'(a) = \frac{\partial}{\partial x} f(a, b)$, which exists by assumption, so

$$f_x(a, b) = g'(a) = 0.$$

• Similarly, $f_y(a, b) = 0$.



Definition

An interior point (a, b) in the domain of a function f is called a **critical point** of f if either $\nabla f(a, b) = \vec{0}$ or at least one of $f_x(a, b)$ and $f_y(a, b)$ does not exist.

Another way to state the first derivative test is that if f has

a local extremum at an interior point P_0 of the domain, then P_0 is a critical point. The converse is not true.

e.g. For $f(x,y) = y^2 - x^2$, $(0,0)$ is a critical point, but it does not give a local extremum.

Def: If all partial derivative of f are zero at an interior point (a,b) of the domain, the (a,b) is called a **saddle point** of f , and $(a,b, f(a,b))$ is called a **saddle point** of the surface $z = f(x,y)$.

Theorem (Second Derivative Test)

Let f be a function whose second partial derivatives are all continuous on a open ~~ball~~^{disk} centered at (a,b) . Suppose that $\nabla f(a,b) = \vec{0}$ (so (a,b) is a critical point of f). Let

$$H := H(a,b) := f_{xx}(a,b)f_{yy}(a,b) - (f_{xy}(a,b))^2.$$

- ▶ If $H > 0$ and $f_{xx}(a,b) > 0$, then f has a local minimum at (a,b) .
- ▶ If $H > 0$ and $f_{xx}(a,b) < 0$, then f has a local maximum at (a,b) .
- ▶ If $H < 0$, then f has no local extremum at (a,b) ; that is, (a,b) is a saddle point ~~of~~ f .

We postpone the proof to a later section.

Remark

If $H = 0$, the second derivative test gives no information: (a, b) could give a local maximum or a local minimum, or it could be a saddle point.

Example

Find all local extrema of the function

$$f(x, y) := x^4 + y^4 - 4xy + 1$$

defined on $\mathbb{R}^2$.

Sol: $(1, 1)$ and $(-1, -1)$, both give a local minimum.

Global/Absolute Extrema

Definition

Let $(a_1, \dots, a_n)$ be a point in the domain D of a function f .

- We say that f has a **global maximum** (or an **absolute maximum**) at $(a_1, \dots, a_n)$ if

$$f(a_1, \dots, a_n) \geq f(x_1, \dots, x_n)$$

for every $(x_1, \dots, x_n) \in D$.

- We say that f has a **global minimum** (or an **absolute minimum**) at $(a_1, \dots, a_n)$ if

$$f(a_1, \dots, a_n) \leq f(x_1, \dots, x_n)$$

for every $(x_1, \dots, x_n) \in D$.

We saw in MAT1001 that a one-variable continuous function defined on a closed and bounded interval must attain its global extrema. The following is an analogous theorem for two-variable functions.

Theorem

Let $f : D \rightarrow \mathbb{R}$ be a continuous function, where D is a closed and bounded set in $\mathbb{R}^2$. Then f has a global maximum $f(x_1, y_1)$ and a global minimum $f(x_2, y_2)$ at some (x_1, y_1) and (x_2, y_2) in D .

Remark: D may not contain the absolute (global) extrema if it is not closed, even if it is bounded. e.g.
 $f(x, y) = x$, $D = \{(x, y) : x^2 + y^2 < 1\}$.

Since, by definition, a global maximum (or minimum) must be ~~use~~ a local maximum (or minimum), the following is a strategy for finding the global extrema of a continuous function defined a closed and bounded domain D .

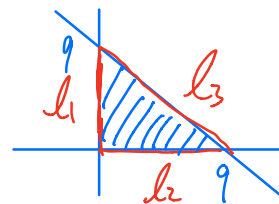
1. Find the values of f at the critical points (in the interior of the domain).
all candidates for local extrema in interior of D
2. Find the local extrema of f on the boundary of D .
3. The largest value and the smallest value from step 1 and 2 are, respectively, the global maximum of f and the global minimum of f .

Example

Find the global maximum and global minimum of the function

$$f(x, y) := 2 + 2x + 4y - x^2 - y^2 \quad \leftarrow \text{cts}$$

defined on the triangular region in the first quadrant bounded by the lines $x = 0$, $y = 0$ and $y = 9 - x$.



Sol (Outline):

• $f_x = 2 - 2x$, $f_y = 4 - 2y$;

$f_x = 0 = f_y \Leftrightarrow (x, y) = (1, 2)$; this is in interior of D .

• Boundary is $\mathcal{L}_1 \cup \mathcal{L}_2 \cup \mathcal{L}_3$, where

$\mathcal{L}_1 := \{(0, y) : 0 \leq y \leq 9\}$, $\mathcal{L}_2 := \{(x, 0) : 0 \leq x \leq 9\}$,

$\mathcal{L}_3 := \{(x, 9-x) : 0 \leq x \leq 9\}$.

• On $\mathcal{L}_3$: $f(x, y) = f(x, 9-x) = -43 + 6x - 2x^2 = p(x)$;

$p'(x) = 0 \Leftrightarrow x = 4 \Leftrightarrow y = 5$, so

$(x, y) = (4, 5)$ is a candidate.

At end points $x=0$ and $x=9$, we have $(0, 9)$ and

$(9, 0)$ also being candidates.

• By doing similar arithmetic for $\mathcal{L}_1$ and $\mathcal{L}_2$, we found the following candidates:

$\mathcal{L}_1$: $(0, 0)$, $(0, 2)$, $(0, 9)$.

$\mathcal{L}_2$: $(0, 0)$, $(1, 0)$, $(9, 0)$.

• Compare all candidates:

$\text{Max } f(1, 2) = 7$, $f(0, 0) = 2$, $f(0, 2) = 6$, $f(0, 9) = -43$,
 $f(1, 0) = 3$, $\text{Min } f(9, 0) = -61$, $f(4, 5) = -11$.

